

Ideas Parked — Not Yet Merged (v15)

Staging document; each entry marks its eventual destination

Two conversations produced one finding, not two. A density/energy pair built from the same per-vector quantity (S_{atom} and $\mathcal{E}_{\text{atom}}$, both sums of r_i transformed) turned out to be forced into a one-directional bound. A second pair, density redefined as bare count and energy redefined as a shell-weighted strength distribution, turned out to have no relationship at all, in either direction. This entry states the rule both results are instances of, gives a reusable test for any future pair of atom-level metrics, and sketches where the distinction might actually matter — all of it exploratory, none of it adopted.

1 The rule

Call a metric on a Collapsed atom *separable* if it is a sum over the atom's own vectors of some fixed per-vector function: $M = \sum_{i=1}^N f(r_i)$. Every metric discussed so far is of this form, including the ones that do not look it: N itself is $\sum_i 1$, the constant function.

The test. For two separable metrics $M_1 = \sum f(r_i)$ and $M_2 = \sum g(r_i)$: if $f(r) \leq g(r)$ for every $r \in [0, 1]$, then $M_1 \leq M_2$ for every atom, with no exceptions — summing a pointwise inequality preserves it. If neither $f \leq g$ nor $f \geq g$ holds everywhere on $[0, 1]$ — if the two functions cross — no such bound exists in either direction, and the two metrics are free to move independently, as far apart as the atom's own vectors allow.

Checked against both findings on file. S_{atom} and $\mathcal{E}_{\text{atom}}$ use $f(r) = r$ and $g(r) = r^2$; checked directly, $r^2 \leq r$ holds across all of $[0, 1]$, so $\mathcal{E}_{\text{atom}} \leq S_{\text{atom}}$ is forced, matching the identity already found. N and the shell-weighted energy sketch use $f(r) = 1$ and $g(r) = \lfloor r/w \rfloor \cdot r$; checked directly, neither $f \leq g$ nor $f \geq g$ holds on all of $[0, 1]$ (the constant function sits below g near $r = 1$ and above it near $r = 0$), so no bound exists, matching the unbounded result already found. One test, both results, no coincidence.

Why this is worth having as a standing test. Any future proposal for a new atom-level metric can be checked against every metric already on file in one step — write both as f and g on $[0, 1]$, check whether the two curves cross — rather than re-running numerical trials each time a new pair is proposed. A crossing means the pair is informative together (two independent readings of the atom); no crossing means one is redundant given the other, except for the gap between them, which is its own quantity (below).

2 The gap, where one exists, is itself a quantity

Where a bound holds, $M_2 - M_1 = \sum (g(r_i) - f(r_i))$ is not nothing — it is a separable metric in its own right, built from the per-vector function $g - f$. For $S_{\text{atom}} - \mathcal{E}_{\text{atom}}$, that function is $r - r^2$,

maximized at $r = 0.5$ and zero at both $r = 0$ and $r = 1$ (already found). The general pattern: wherever two metrics are forced into a bound, the slack between them measures something like efficiency — how much of one quantity a vector contributes without contributing a matching share of the other — and is itself worth naming once a specific bounded pair is adopted, not only in this one case.

3 Where this might be used, if any of it is adopted

None of this is proposed as settled; each is a direction worth testing, not a result.

A minimal, non-redundant reporting set. With seven metrics already on file and at least two more count-based ones proposed, not all of them are worth reporting side by side. The test above sorts the existing inventory into two kinds: pairs that are bounded (reporting both is partly redundant — the gap is the real second number) and pairs that are free (reporting both is two genuinely independent readings). A table applying the test to every pair already in the inventory — S_{atom} , V_{atom} , γ_{atom} , $\mathcal{E}_{\text{atom}}$, N , D — would say, once, which combinations are worth stating together.

A two-axis placement for comparing atoms. Where a pair is free (density and shell-energy, as checked here), plotting a population of Collapsed atoms on those two axes is a meaningful scatter, not a degenerate one — nothing forces the points onto or under a curve the way a bounded pair would. Where a pair is bounded instead (S_{atom} against $\mathcal{E}_{\text{atom}}$), the same plot is still informative, but differently: every atom must fall on or below the line $\mathcal{E} = S$, and *how far below* is the efficiency reading from Section 2, not a free coordinate.

A possible reading for Inference’s own evidence questions, offered cautiously. The common split in evidence aggregation — many weak signals versus one strong one — is roughly what a free density/energy pair would distinguish: an atom that is dense but low-energy under the shell reading looks like many weak contributions; one that is light but high-energy looks like a single strong one. This is an analogy to be tested against Inference’s own existing reference-atom mechanism, not a replacement for it — it may turn out to duplicate a distinction that mechanism already makes some other way, which would need checking before anything is built on it.

A post-hoc description of Assembly’s frozen camps, not a new input to them. Section 5 of ideas_v13 closed the question of feeding any density- or mass-like weighting into Assembly’s own motion step, on direct, tested evidence that it breaks monotonic convergence. That finding is not reopened here. What is not closed: describing a camp’s own resulting density and energy, once frozen, as a label attached after the fact — characterizing what rest produced, rather than steering how it got there. Whether this descriptive label would be useful for anything downstream is not examined here.

A check on construction, not a claim about meaning. If a later paper’s pipeline is expected, empirically, to produce atoms within some typical density/energy relationship, an atom that falls far outside it — for a free pair, there is no “outside”; for a bounded pair, violating the bound at all would be impossible by construction and would indicate an error in how the atom was built, not a new kind of atom. This is a debugging use, not an analytical one, but worth keeping in

mind precisely because a bounded pair makes some configurations provably unreachable rather than merely unlikely.

4 Status

The rule and its test are checked and stand on their own, independent of any naming question. The four uses above are genuinely speculative, offered as directions worth trying rather than conclusions — in particular, the Inference analogy needs checking against that paper’s own machinery before it is anything more than a resemblance, and the debugging use depends on a bounded pair actually being adopted into the formal inventory first.